Every train to spec

Mine-to-port grade compliance for Fortescue, on Databricks

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The outcome: grade known at departure, not 36 hours later

Solomon trains below typical Fe

73% today

→ under 5% (simulated)

by blending on lab-reconciled grade

Off-spec trains caught at load-out

11% → 71%

flagged at departure, 24-48 h before the lab result

Grade penalties avoided

US$5.6M/yr at risk (illustrative)

Planning case: US$2.8M/yr

Why trains miss spec today

- Controllers build each ~34.5 kt train from block-model stockpile grades
- The block model is biased: Solomon over-states Fe by ~0.45%, Cloudbreak by ~0.22%
- The cross-belt analyser drifts between calibrations
- The lab assay, the grade that counts, arrives 24-48 h after departure
- By then the ore is at Herb Elliott Port: re-blend, rehandle or pay the penalty
- Result: 39% of trains below typical Fe; hubs that over-deliver give iron away

For the executive sponsor

GM Integrated Planning & Marketing

Fewer off-spec vessels: 16 of 76 below typical → 0 (simulated)

US$2.8M/yr planning case, US$5.6M/yr upside

One certified definition of compliance

Auditable decisions for every train

What we ask

8-week pilot at Solomon, the problem hub

Real LIMS, FMS and analyser feeds

Success = below-typical trains halved

Go/no-go on value, not technology

For the domain owner: grade control

Superintendent Grade Control

Predicted lab grade for every in-flight train

Recommended blend for every planned train

One click to approve, re-plan or reject

Outage? Re-plan in seconds without that stockpile

Why you can trust it

Fe error 0.249% → 0.111% vs block model

Accurate at every hub: 0.11% MAE at Solomon

Shows the block-model view next to each blend

Controllers stay in charge; overrides are logged

One integrated data journey

- Lakeflow: FMS hauls, rail, analyser, LIMS and port feeds ingested; quality rules drop bad assays
- Unity Catalog: hub-level row filters, masked penalty $, certified metrics, lineage
- ML + GenAI: train-grade model (MLflow, UC champion), blend optimiser, AI rationale
- Lakebase: live stockpile, prediction and blend state; controller decisions written back
- Genie Agent: ask compliance questions in plain English, governed by the same rules
- Databricks App: the Blend Control Room for every shift

The model: knowing the lab grade before the lab

Signal available at departure	Fe error (MAE)	Below-typical trains caught	False alarms
Block model (today)	0.249%	11%	1
Cross-belt analyser (raw)	0.172%	66%	20
Train-grade model (Databricks)	0.111%	71%	14

Value model: 30-day backtest, annualised

KPI	Today	With Databricks (simulated)
Trains below typical Fe	39.2%	5.9%
Solomon trains below typical Fe	72.7%	0.8%
Vessels below typical Fe (30 d)	16 of 76	0 of 76
Vessel grade penalty, annualised	US$5.6M	~US$0
Planning case (50% realisation)		US$2.8M/yr avoided
Trade-off: avg train Fe	57.82%	58.07% (more high-grade ore used)

Governed by design

- A Solomon grade controller sees only Solomon trains (UC row filter)
- Penalty dollars are visible to commercial roles only (UC column mask)
- Genie and the app inherit the same rules; no separate permission model
- Certified metric views: one definition of 'below typical'
- Every recommendation, override and reason is logged and lands in Delta
- Full lineage from raw feed to the app's Lakebase tables

Next steps: 8-week Solomon pilot

- Weeks 1-2: connect LIMS, FMS, analyser and rail feeds through Lakeflow Connect
- Weeks 3-4: retrain on Fortescue history; agree penalty terms and certified KPIs
- Weeks 5-7: shadow mode in the Blend Control Room alongside current practice
- Week 8: value review against the baseline; go/no-go to all hubs
- Success criteria: below-typical trains halved at Solomon; no extra controller effort



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